

Clue Selection Under Noise

Codenames spymaster, baseline model

1 Setup

Let A be the set of board words we want the guesser to select and B the set we want them to avoid. A clue is a clue word together with a number k , and announcing it allows the guesser k selections. Fix a clue word, and let a_i be its similarity to word $i \in A$ and b_w its similarity to word $w \in B$, where the a_i are sorted so that $a_1 \geq a_2 \geq \dots$. Selecting a word of A is worth 1, and selecting $w \in B$ costs $c_w > 0$. We want to choose both parts of the clue.

We assume the guesser perceives word x 's similarity with the clue word as $\hat{z}_x = z_x + \varepsilon_x$, where z_x is our similarity to x and the ε_x are independent $N(0, \sigma^2)$. This models differences between our interpreted embeddings and the guesser's — higher σ means a larger gap between the two. We also assume the guesser selects unrevealed words in decreasing order of $\hat{z}$, stopping after k selections or upon selecting a word of B , whichever comes first.

In words, the guesser ranks the board the same way we do, perturbed by independent noise of a single scale, and then guesses greedily. σ is the only free parameter.

2 Reduction

Let

$$D = \max_{w \in B} \hat{z}_w, \quad W^* = \arg \max_{w \in B} \hat{z}_w, \quad N = \#\{i \in A : \hat{z}_i > D\}.$$

Every word ranked above D is in A , since D is the maximum of $\hat{z}$ over B . Thus the guesser's ordering begins with the N words counted above, and the $(N + 1)$ -th word is W^* . This means the guesser reveals $\min(k, N)$ words of A , and selects a word of B if and only if $N < k$, in which case that word is W^* .

Note that no word of B other than W^* can be selected, so the single random variable D accounts for all of B . Write φ and Φ for the density and the distribution function of a standard normal. Each $\hat{z}_w$ is $N(b_w, \sigma^2)$, so

$$F_{\hat{z}_w}(x) = \Phi\left(\frac{x - b_w}{\sigma}\right), \quad f_{\hat{z}_w}(x) = \sigma^{-1} \varphi\left(\frac{x - b_w}{\sigma}\right).$$

Since D is a maximum of independent variables,

$$F_D(d) = \prod_{w \in B} F_{\hat{z}_w}(d) = \prod_{w \in B} \Phi\left(\frac{d - b_w}{\sigma}\right).$$

3 Conditional distribution

D is a function of $(\varepsilon_w)_{w \in B}$ only, which is independent of $(\varepsilon_i)_{i \in A}$. Thus conditioning on $D = d$ does not change the distribution of the $\hat{z}_i$, and each $i \in A$ independently satisfies $\hat{z}_i > d$ with probability

$$p_i(d) = \Phi\left(\frac{a_i - d}{\sigma}\right).$$

Given $D = d$, the count N is therefore a sum of $|A|$ independent Bernoulli variables, the i -th with success probability $p_i(d)$.

4 Objective

From Section 2 the guesser reveals $\min(k, N)$ words of A , and selects W^* exactly when $N < k$. Each word of A is worth 1 and W^* costs c_{W^*} , so the value of announcing k is $G(k) - P(k)$, where

$$G(k) = \mathbb{E}[\min(k, N)], \quad P(k) = \mathbb{E}[c_{W^*} \mathbf{1}\{N < k\}].$$

Since $\min(k, N) = \sum_{j=1}^k \mathbf{1}\{N \geq j\}$, taking expectations and conditioning on D gives

$$G(k) = \sum_{j=1}^k \int \Pr(N \geq j \mid D = d) dF_D(d).$$

Each $\Pr(N \geq j \mid D = d)$ follows from the Bernoulli sum of Section 3, and the integral is one-dimensional in d . Section 5 computes both. It remains to find P .

In P the cost c_{W^*} and the event $\{N < k\}$ both depend on the noise, but they become independent once D is fixed. To see this, note that D is a function of $(\varepsilon_w)_{w \in B}$ alone, so conditioning on $D = d$ constrains that family and leaves $(\varepsilon_i)_{i \in A}$ with its original distribution. Given $D = d$, the count $N = \#\{i \in A : z_i + \varepsilon_i > d\}$ is a function of $(\varepsilon_i)_{i \in A}$, while the identity W^* is a function of $(\varepsilon_w)_{w \in B}$. The two families are independent, so N and W^* are conditionally independent given D , and the expectation splits:

$$P(k) = \int \Pr(N < k \mid D = d) \bar{c}(d) dF_D(d), \quad \bar{c}(d) = \mathbb{E}[c_{W^*} \mid D = d].$$

The first factor is again a probability for the Bernoulli sum of Section 3. It remains to compute $\bar{c}(d)$, for which we need $\Pr(W^* = w \mid D = d)$.

Writing g_w for the joint density of $W^* = w$ and D at d ,

$$\Pr(W^* = w \mid D = d) = \frac{g_w(d)}{f_D(d)}.$$

$W^* = w$ and $D = d$ means $\hat{z}_w = d$ and $\hat{z}_v < d$ for every other $v \in B$. The $\hat{z}$ are independent, so

$$g_w(d) = f_{\hat{z}_w}(d) \prod_{v \neq w} F_{\hat{z}_v}(d).$$

The events $\{W^* = w\}$, $w \in B$, are disjoint and exhaustive, so

$$f_D(d) = \sum_{w \in B} g_w(d).$$

From Section 2, $F_D(d) = \prod_{v \in B} F_{\hat{z}_v}(d)$, so the product over $v \neq w$ is $F_D(d)/F_{\hat{z}_w}(d)$. Set

$$h_w(d) = \frac{f_{\hat{z}_w}(d)}{F_{\hat{z}_w}(d)} = \frac{\sigma^{-1} \varphi((d - b_w)/\sigma)}{\Phi((d - b_w)/\sigma)},$$

so that $g_w(d) = F_D(d) h_w(d)$ and $f_D(d) = F_D(d) \sum_v h_v(d)$. The $F_D(d)$ cancels in the ratio:

$$\Pr(W^* = w \mid D = d) = \frac{h_w(d)}{\sum_v h_v(d)}, \quad \bar{c}(d) = \frac{\sum_w c_w h_w(d)}{\sum_w h_w(d)}.$$

We choose the clue maximizing the value,

$$\text{clue}^* = \arg \max_{\text{word}, 1 \leq k \leq K} [G(k) - P(k)].$$

5 Computation

Both G and P have the form $\int \psi(d) dF_D(d)$ for a bounded ψ :

$$\psi_G(d) = \sum_{j=1}^k \Pr(N \geq j \mid D = d), \quad \psi_P(d) = \Pr(N < k \mid D = d) \bar{c}(d).$$

Both are evaluated by numerical integration. Partition an interval $[d_0, d_M]$ into cells $[d_m, d_{m+1}]$, take ψ at the midpoint $\bar{d}_m = \frac{1}{2}(d_m + d_{m+1})$ of each, and weight it by the probability that D lands in that cell:

$$u_m = F_D(d_{m+1}) - F_D(d_m), \quad \int \psi(d) dF_D(d) \approx \sum_m \psi(\bar{d}_m) u_m.$$

The u_m are exact, being differences of the product in Section 2, so the only error is the variation of ψ within a cell.

Take d_0 below every b_w and d_M above every b_w , each by several σ , so that $F_D(d_0)$ and $1 - F_D(d_M)$ are negligible and no mass falls outside the grid. The same grid and the same u_m serve both integrals, so one pass over the cells gives $G(k)$ and $P(k)$ together, and for every $k \leq K$ at once.

Within a cell, $\bar{c}(\bar{d}_m)$ is immediate from its definition, at a cost of $|B|$ terms. The probabilities $\Pr(N \geq j \mid D = \bar{d}_m)$ follow from the standard recursion for a sum of independent Bernoulli variables. Letting $q_r^{(i)}$ be $\Pr(N = r)$ after the first i words of A ,

$$q_0^{(0)} = 1, \quad q_r^{(i)} = q_r^{(i-1)}(1 - p_i(\bar{d}_m)) + q_{r-1}^{(i-1)} p_i(\bar{d}_m),$$

and $\Pr(N \geq j \mid D = \bar{d}_m) = 1 - \sum_{r < j} q_r$. Counts above K are never used, so the recursion may be truncated there, costing $|A|K$ per cell rather than $|A|^2$.

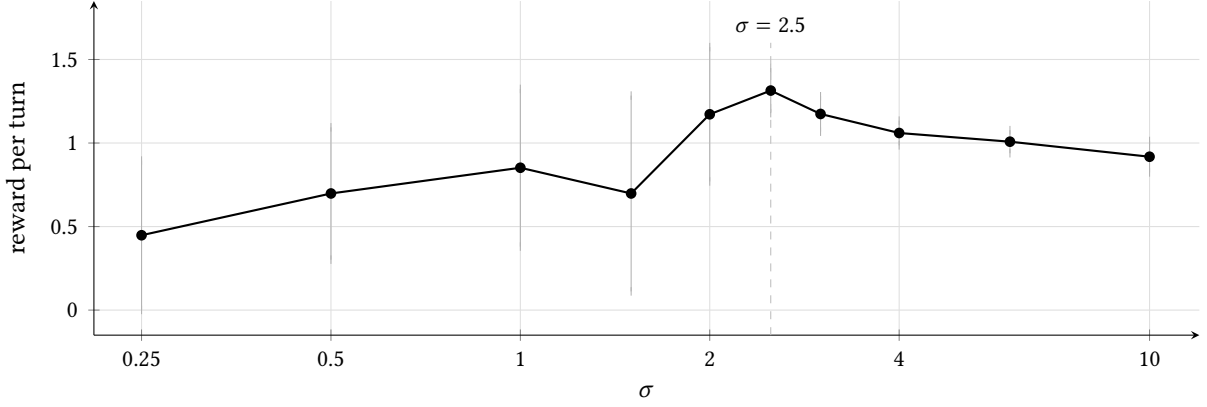
The total is $O(M(|A|K + |B|))$ per clue, with M the number of cells. Every step is a fixed sequence of arithmetic on arrays of the same shape, so the whole clue vocabulary is scored at once rather than one clue at a time.

6 Results

The model above has one free parameter. To choose it, σ was swept over three orders of magnitude and each value was made to play the same 100 positions, drawn from held-out board words and spanning openings through endgames. Each turn was scored as the game scores it: the guesser's ranking is walked, own words are counted until the first word of B , and the reward is +1 per own word with -0.2 , -1 and -10 for a neutral, opponent and assassin word respectively. The guesser was an external language model, not the similarity ranking the spymaster itself uses, so the numbers measure the model against a listener it has no access to.

σ	announced	revealed	gap	reward	s.e.	asn	opp	neu	clean
0.25	3.96	1.55	-2.41	0.448	0.228	5	53	36	6
0.5	3.91	1.60	-2.31	0.698	0.202	3	53	36	8
1.0	3.45	1.74	-1.71	0.852	0.241	4	43	29	24
1.5	2.79	1.72	-1.07	0.698	0.299	7	28	21	44
2.0	2.33	1.69	-0.64	1.172	0.205	3	19	14	64
2.5	1.72	1.41	-0.31	1.314	0.069	0	7	13	80
3.0	1.40	1.22	-0.18	1.174	0.054	0	3	8	89
4.0	1.21	1.10	-0.11	1.060	0.038	0	3	5	92
6.0	1.16	1.05	-0.11	1.008	0.035	0	3	6	91
10.0	1.13	0.99	-0.14	0.918	0.048	0	6	6	88

Per turn, over 100 positions. *announced* is k ; *revealed* is the number of words of A the guesser actually took; the last four columns count how each turn ended.



Realized reward per turn. Bars are 95% confidence intervals; the wide bars at small σ are assassin turns, each costing -10 .

Both limits saturate. As $\sigma \rightarrow 0$ the model treats the guesser as a copy of itself and announces $k = K$ on nearly every board, and as σ grows it announces $k = 1$ and stops responding to further increases. Neither end is optimal, and the reward curve is single-peaked between them.

Note that the number of words revealed and the reward earned are maximized at different σ . Revealed words peak at $\sigma = 1.0$ with 1.74 per turn, where reward is only 0.852; reward peaks at $\sigma = 2.5$ with 1.314, where 1.41 words are revealed. The difference is entirely risk. At $\sigma = 1.0$ the guesser reaches a word of *B* on 76 of 100 turns and the assassin on 4; at $\sigma = 2.5$ it reaches *B* on 20 and the assassin on none. Thus the gain from claiming an extra word is real but is more than repaid, and an objective fitted on revealed words alone would choose a value that loses games.

The assassin is what makes the left half of the curve hard to measure. It is worth -10 , so a handful of assassin turns dominate the variance: the standard error is 0.2 to 0.3 for $\sigma \leq 2.0$ and 0.035 to 0.069 for $\sigma \geq 2.5$, where no assassin was reached at all. Comparing paired by board, $\sigma = 2.5$ beats every other value tested at the 5% level except $\sigma = 2.0$ ($+0.14 \pm 0.21$) and $\sigma = 1.0$ ($+0.46 \pm 0.49$), both of which are indistinguishable only because their own assassin losses are so noisy.

Note also that the gap between announced and revealed is negative everywhere and closes monotonically, reaching -0.11 at $\sigma = 4$. The model therefore remains optimistic at its own optimum, announcing 1.72 where 1.41 is delivered. Calibrating G and maximizing reward are different objectives: only the ordering of $G(k) - P(k)$ across clues and k affects the choice made, so a bias shared by every candidate leaves the argmax unchanged. Fitting σ to close the gap would report $\sigma \approx 4$ and cost 0.25 reward per turn.

We therefore take $\sigma = 2.5$. The positions used here are disjoint from the evaluation boards, so the value was not chosen on the boards it is later reported on.

Here a_i and b_w are similarities standardized per clue, c_w are the costs of the words to avoid, and σ is the guesser's per-word perturbation.